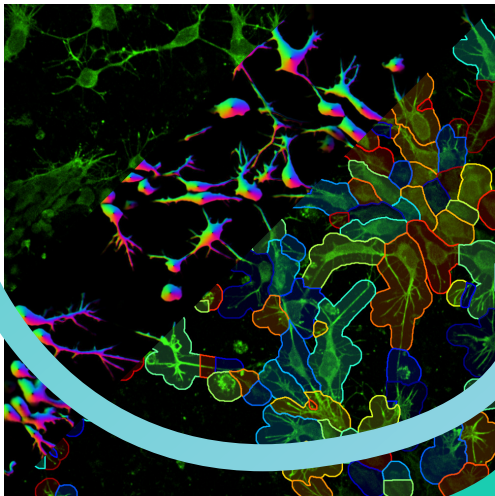


JUNE 2, 2026

Cellpose for machine learning in image analysis

Pasteur-BiolImage-Analysis-Course-2026
Julie Mabon
Biological Analysis Unit - Institut Pasteur



summary

01. INSTANCE SEGMENTATION

02. FEATURE EXTRACTION WITH CONVOLUTIONAL NEURAL NETWORKS

03. CONVOLUTIONAL NEURAL NETWORKS FOR INSTANCE SEGMENTATION

04. PRACTICAL

PART 1

Instance Segmentation

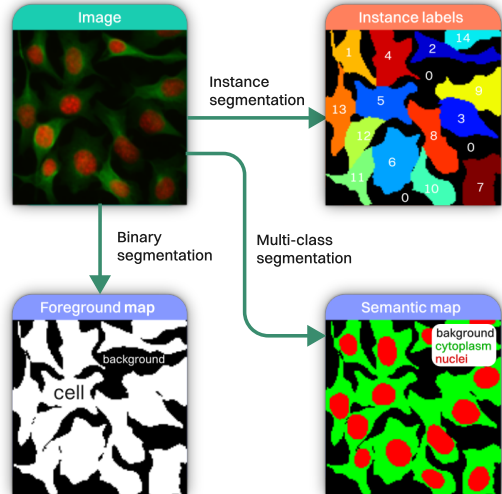
Instance Segmentation

Semantic Segmentation

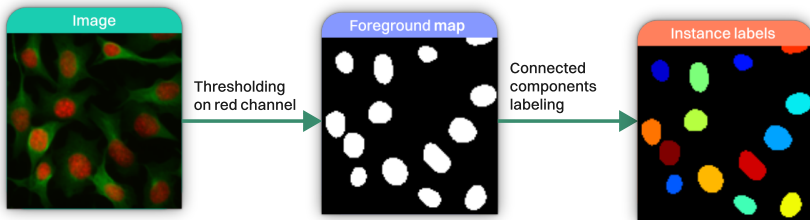
- Assigning a class/category to each pixel
- Binary : foreground/background
- Multiclass : cytoplasm/nuclei/bg

Instance Segmentation

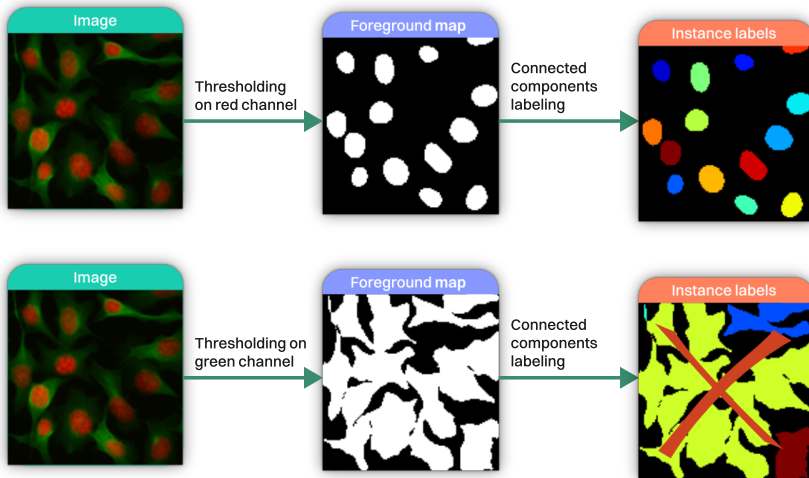
- Separating individual cells



The old reliable (?) method



The old reliable (?) method

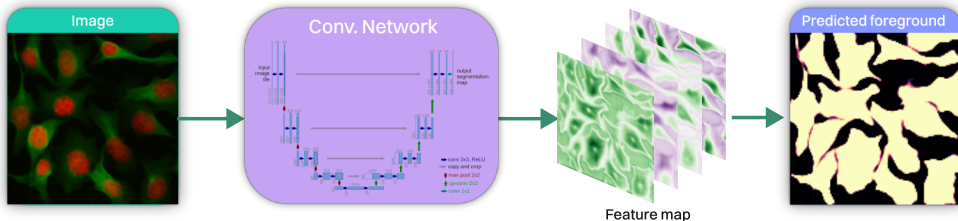


PART 2

Feature extraction with Convolutional Neural Networks

Convolutional Neural Networks: Semantic segmentation

- **Convolutional Neural Networks** learn to extract **features** from images
- Relevant information is extracted from the **features**
- Image to image **translation** / **Local** by construction



Convolutional Neural Networks: Training

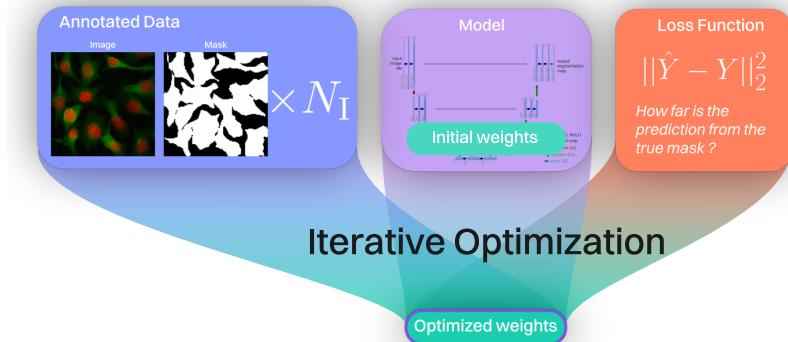
Training

Estimating the best weights given a bunch of **annotated data**, a **model** and a **loss function** (measure of the prediction error).

Convolutional Neural Networks: Training

Training

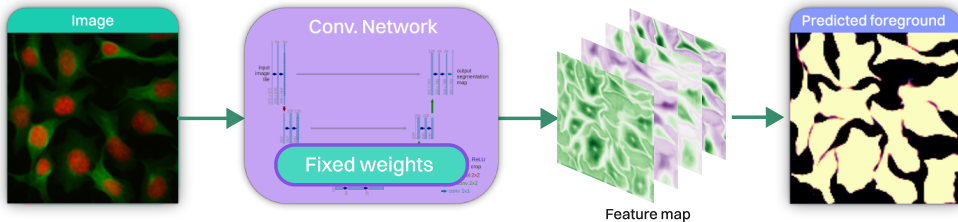
Estimating the best weights given a bunch of **annotated data**, a **model** and a **loss function** (measure of the prediction error).



Convolutional Neural Networks: Evaluation

Evaluation

Passing an image through the model with **fixed** weights/parameters.



Fine tuning

- A model is generally good on the same **domain** (eg similar looking images) that it was trained on
- We have a great **diversity** of imaging modalities / imaged samples
- You may need to **tune** the model to your specific task

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Fine-tuning

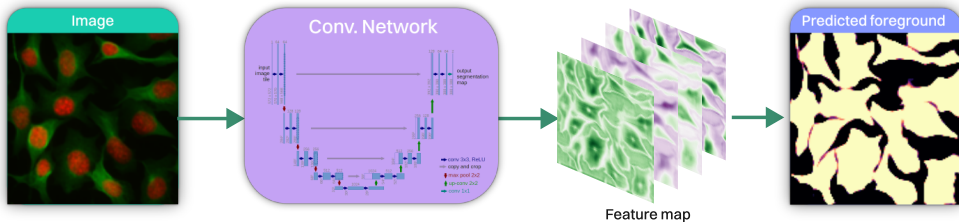
Further optimizing a **pre-trained** model on a **specific** task dataset.

PART 3

Convolutional Neural Networks for Instance segmentation

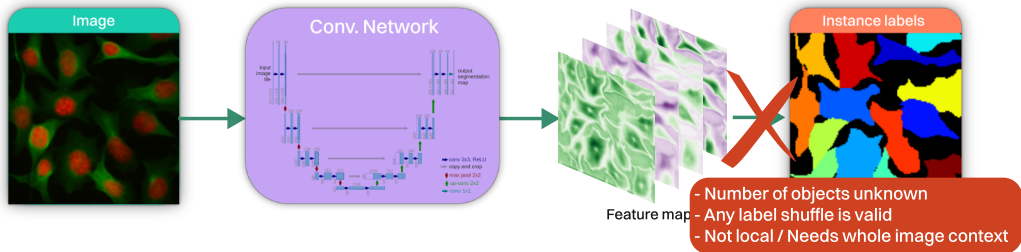
Convolutional Neural Networks: Instance segmentation

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Convolutional Neural Networks: Instance segmentation

- Convolutional Neural Networks learn to extract features from images
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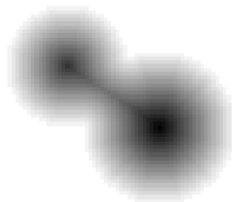


Watershed

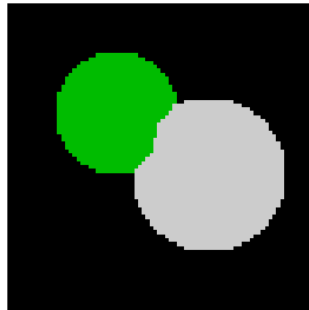
Overlapping objects



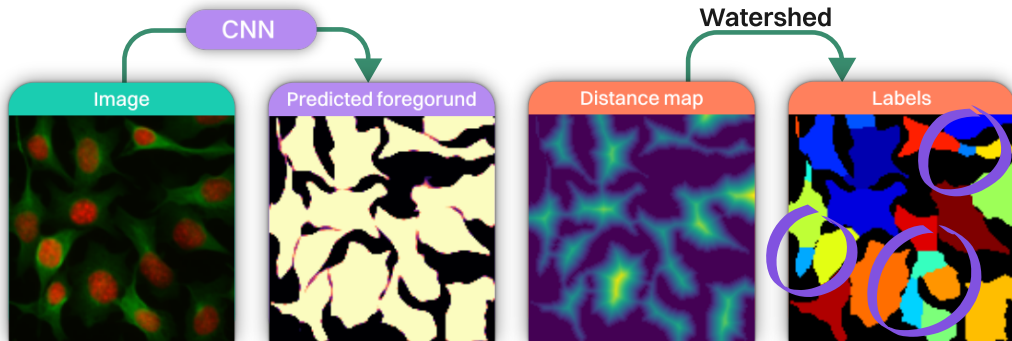
Distances



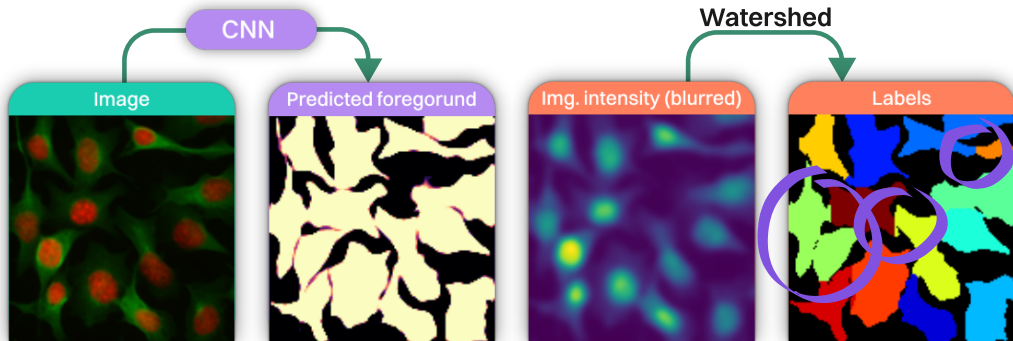
Separated objects



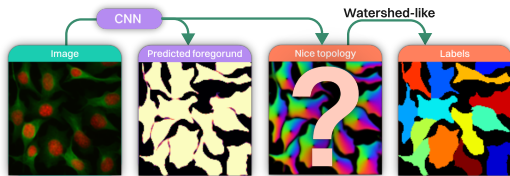
Watershed



Watershed



From watershed to cellpose



Watershed limitations

- Simple distance map may fail
- Image intensity can be inhomogeneous

Results

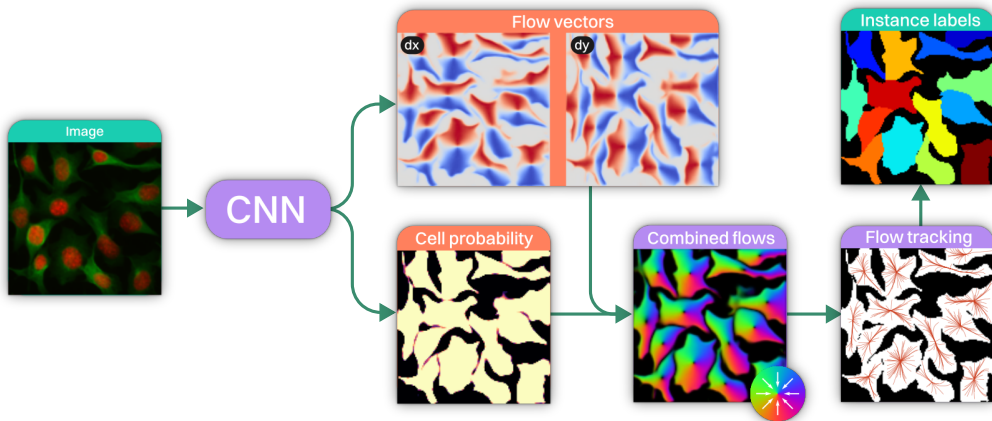
Model design. Classical segmentation approaches based on the watershed algorithm^{28,29} use the grayscale values of an image to create a topological map, whose basins represent the segmented regions. These methods work well when the segmented objects have intensity profiles that decay smoothly from the center, so that the watershed forms a single basin. However, many types of cell form multiple intensity basins, usually due to the nuclear exclusion of the fluorescent marker and its inhomogeneous distribution along cell borders and protuberances. These issues motivated us to construct an intermediate representation of an object that does form a single smooth topological basin (Fig. 1a,b).

We generated topological maps through a process of simulated diffusion that used the ground-truth masks drawn by a human annotator (Fig. 1a). A neural network was then trained to predict the horizontal and vertical gradients of the topological maps, as well as a binary map that indicates if a given pixel is inside or outside of regions of interest (ROI) (Fig. 1c,d). On a test image, the neural network predicted the horizontal and vertical gradients, which formed vector fields. By following the vector fields through a process known as gradient tracking²⁹, all pixels belonging to a given cell can be routed to its center. Thus, by grouping together pixels that

C. Stringer et al., "Cellpose: a generalist algorithm for cellular

segmentation," *Nat Methods*, vol. 18, no. 1, pp. 100–106, Jan. 2021

Cellpose: Instance segmentation as flow tracking



Cellpose: Instance segmentation as flow tracking

Cellpose in short

- For **cytoplasm** and **nuclei** instance segmentation
- Trained on microscopy **fluorescence** images
- Limited to **blob-like** & **non-overlapping** cells
- Based on a lightweight **Convolutional Neural Network** $\implies$ easy to fine-tune
True for cellpose 3. Cellpose-SAM is much more computationally demanding.

PART 4

Practical

Practical

On the github go to
[ECI/Day2_Cellpose/Day2_Cellpose.md](#)

Program

1. Using the Cellpose plugin in Fiji
2. Fitting the parameters to the image
3. Fine-tuning our own models



Cellpose for machine learning in image analysis

Instructor: Julie Mabon, Post-doc, Unité d'Analyse d'Images Biologiques, Institut Pasteur

Note

In this course we use Cellpose 3.0 instead of the latest Cellpose-SAM as it is faster to run on any laptop and easier to re-train.



Getting ready

Warning

Be sure to have all tools installed as instructed in [Installation](#)



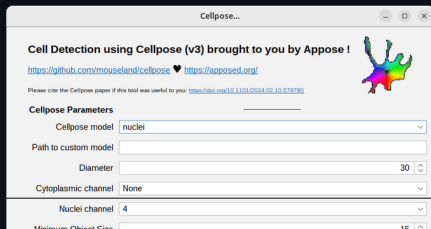
Part 1: Cellpose-Apouse on Fiji

Use case: Segment nuclei in the image, and count spot intensity in each nucleus, then export a CSV using the [Cellpose - Apouse Fiji plugin](#).

1. Load the image
2. Launch the Cellpose plugin:
 - Plugins > Segmentation > Cellpose-Apouse > Cellpose...
3. Set up parameters as suggested. You can hover your mouse over parameters to see what they do.

Important

- Choose appropriate **Cellpose Model**: here we use **nuclei** since we are segmenting nuclei; for cytoplasm use **cyto3**.
- Set **Cytoplasmic/Nuclei channel** according to your image: here we don't have a cytoplasmic channel (→ None), and the nuclei staining is in channel 4
- Make sure to check **return ROIs** to get ROIs in Fiji for our measurements



Cellpose flow definition

